

OpenStax Calculus Volume 3

Section 7.2 Exercises: Nonhomogeneous Linear Equations

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Solve the following equations using the method of undetermined coefficients.

1. $2y'' - 5y' - 12y = 6$
2. $3y'' + y' - 4y = 8$
3. $y'' - 6y' + 5y = e^{-x}$
4. $y'' + 16y = e^{-2x}$
5. $y'' - 4y = x^2 + 1$
6. $y'' - 4y' + 4y = 8x^2 + 4x$
7. $y'' - 2y' - 3y = \sin 2x$
8. $y'' + 2y' + y = \sin x + \cos x$
9. $y'' + 9y = e^x \cos x$
10. $y'' + y = 3 \sin 2x + x \cos 2x$
11. $y'' + 3y' - 28y = 10e^{4x}$
12. $y'' + 10y' + 25y = xe^{-5x} + 4$

In each of the following problems,

- (a) Write the form for the particular solution $y_p(x)$ for the method of undetermined coefficients.
- (b) [T] Use a computer algebra system to find a particular solution to the given equation.

13. $y'' - y' - y = x + e^{-x}$
14. $y'' - 3y = x^2 - 4x + 11$
15. $y'' - y' - 4y = e^x \cos 3x$
16. $2y'' - y' + y = (x^2 - 5x)e^{-x}$
17. $4y'' + 5y' - 2y = e^{2x} + x \sin x$
18. $y'' - y' - 2y = x^2 e^x \sin x$

Solve the differential equation using either the method of undetermined coefficients or the variation of parameters.

19. $y'' + 3y' - 4y = 2e^x$

20. $y'' + 2y' = e^{3x}$

21. $y'' + 6y' + 9y = e^{-x}$

22. $y'' + 2y' - 8y = 6e^{2x}$

Solve the differential equation using the method of variation of parameters.

23. $4y'' + y = 2 \sin x$

24. $y'' - 9y = 8x$

25. $y'' + y = \sec x, 0 < x < \pi/2$

26. $y'' + 4y = 3 \csc 2x, 0 < x < \pi/2$

Find the unique solution satisfying the differential equation and the initial conditions given, where $y_p(x)$ is the particular solution.

27. $y'' - 2y' + y = 12e^x, y_p(x) = 6x^2e^x, y(0) = 6, y'(0) = 0$

28. $y'' - 7y' = 4xe^{7x}, y_p(x) = \frac{2}{7}x^2e^{7x} - \frac{4}{49}xe^{7x}, y(0) = -1, y'(0) = 0$

29. $y'' + y = \cos x - 4 \sin x, y_p(x) = 2x \cos x + \frac{1}{2}x \sin x, y(0) = 8, y'(0) = -4$

30. $y'' - 5y' = e^{5x} + 8e^{-5x}, y_p(x) = \frac{1}{5}xe^{5x} + \frac{4}{25}e^{-5x}, y(0) = -2, y'(0) = 0$

In each of the following problems, two linearly independent solutions— y_1 and y_2 —are given that satisfy the corresponding homogeneous equation. Use the method of variation of parameters to find a particular solution to the given nonhomogeneous equation. Assume $x > 0$ in each exercise.

31. $x^2y'' + 2xy' - 2y = 3x, y_1(x) = x, y_2(x) = x^{-2}$

32. $x^2y'' - 2y = 10x^2 - 1, y_1(x) = x^2, y_2(x) = x^{-1}$